MEDICAL DECISION MAKING : AN INTERNATIONAL JOURNAL OF THE SOCIETY FOR MEDICAL DECISION MAKING • 2021.0

Top 100 Orthopedic AI • Rank #94

Personalization of Medical Treatment Decisions: Simplifying Complex Models while Maintaining Patient Health Outcomes

C. Weyant, M. Brandeau

Department of Management Science and Engineering, Stanford University, Stanford, CA, USA.

Journal:	Publication Year:	Citations:	DOI:
Medical decision making : an international journal of the Society for Medical Decision Making	2021.0	12	10.1177/0272989x211037921

CLINICAL & AGENTIC AI TAKEAWAY

Our machine learning method simplifies complex medical treatment decision models while maintaining patient health outcomes, making them more adoptable and improving equity in patient outcomes.

ABSTRACT

Background: Personalizing medical treatments based on patient-specific risks and preferences can improve patient health. However, models to support personalized treatment decisions are often complex and difficult to interpret, limiting their clinical application. Methods: We present a new method, using machine learning to create meta-models, for simplifying complex models for personalizing medical treatment decisions. We consider simple interpretable models, interpretable ensemble models, and noninterpretable ensemble models. We use variable selection with a penalty for patient-specific risks and/or preferences that are difficult, risky, or costly to obtain. We interpret the meta-models to the extent permitted by their model architectures. We illustrate our method by applying it to simplify a previously developed model for personalized selection of antipsychotic drugs for patients with schizophrenia. Results: The best simplified interpretable, interpretable ensemble, and noninterpretable ensemble models contained at most half the number of patient-specific risks and preferences compared with the original model. The simplified models achieved 60.5% (95% credible interval [crI]: 55.2-65.4), 60.8% (95% crI: 55.5-65.7), and 83.8% (95% crI: 80.8-86.6), respectively, of the net health benefit of the original model (quality-adjusted life-years gained). Important variables in all models were similar and made intuitive sense. Computation time for the meta-models was orders of magnitude less than for the original model. Limitations: The simplified models share the limitations of the original model (e.g., potential biases). Conclusions: Our meta-modeling method is disease- and model- agnostic and can be used to simplify complex models for personalization, allowing for variable selection in addition to improved model interpretability and computational performance. Simplified models may be more likely to be adopted in clinical settings and can help improve equity in patient outcomes.

Keywords: Schizophrenia, Personalized Medicine, Medical Decision Making, Meta-model

Study Overview & Background

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Methodology & Clinical Focus

Study Classification: theoretical, modeling, or simulation study. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

Our machine learning method simplifies complex medical treatment decision models while maintaining patient health outcomes, making them more adoptable and improving equity in patient outcomes.

Extended Repository Abstract

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Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** C. Weyant, M. Brandeau
- **Journal / Venue:** Medical decision making : an international journal of the Society for Medical Decision Making (2021.0)
- **SJR Metric:** Quartile 1.0
- **Citation Count:** 12 citations recorded
- **Digital Object Identifier (DOI):** https://doi.org/10.1177/0272989x211037921
- **Consensus Evidence Link:** https://consensus.app/papers/personalization-of-medical-treatment-decisions-weyant-brandeau/8b389a3b786f5042a2d4909766a6412f/

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